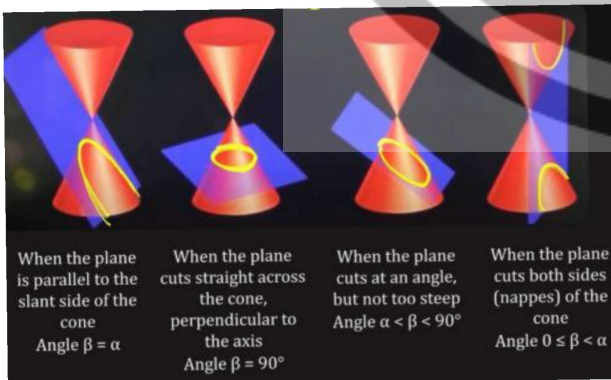
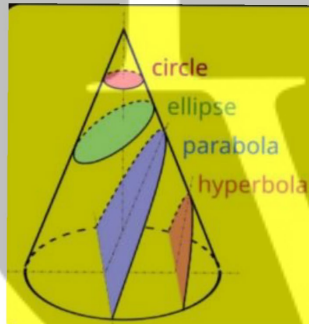
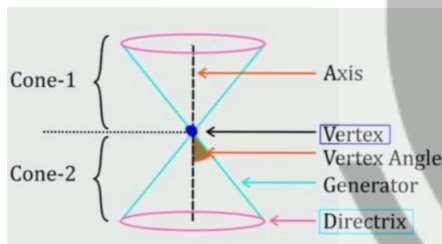


Conic Sections:

- Conic Sections = different shapes formed by slicing a cone in different ways.
- Conic sections are shapes made when a cone is cut by a flat surface.

There are 4 main types:

- Circle - A flat cut straight across the cone.
- Ellipse - A slanted cut that doesn't go through the base
- Parabola - A cut parallel to the side of the cone.
- Hyperbola - A cut that goes through both halves of a double cone.



* Parabola: Jab Plane ko generator ke Parallel cut kare.

* Hyperbola: Jab Plane cone ko double cone ko cut kare aur angle > generator angle ho.

Cases:

i) Circle -

When $\beta = 90^\circ$ (Plane cuts the cone flat across)

Shape: Circle

ii) Ellipse -

When $\alpha < \beta < 90^\circ$ (Plane is slanted but not too steep)

Shape: Ellipse

iii) Parabola -

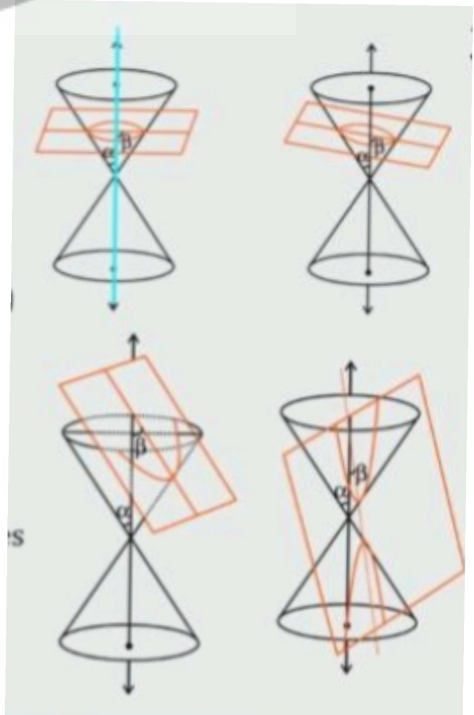
When $\beta = \alpha$ (Plane is Parallel to the cone's side)

Shape: Parabola

iv) Hyperbola -

When $0 \leq \beta < \alpha$ (Plane is very steep, cuts both nappes of cone)

Shape: Hyperbola.



Types of Conic Sections:

* Circle: Jab Plane ko base ke Parallel cut kare.

* Ellipse: Jab Plane cone ko base ke Parallel se thoda tilted cut kare
(but angle < generator Angle)

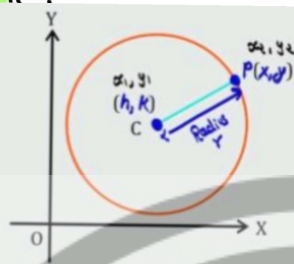
Circle -

A circle is the set of all points in a plane that are at the same distance from a fixed point.

- Fixed Point = Centre of the circle
- Distance from centre to circle = Radius.

Equation of a circle:

If the centre is $C(h, k)$ and the radius is r , By (Distance formula)



$$(x-h)^2 + (y-k)^2 = r^2$$



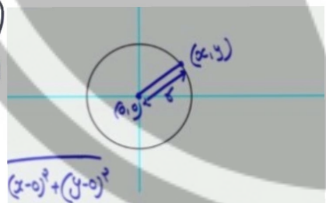
$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Distance PC

$$r = \sqrt{(x-h)^2 + (y-k)^2}$$

Special Case:

If the centre is at the origin $(0, 0)$ Then the equation becomes:



$$x^2 + y^2 = r^2$$

$$(x-h)^2 + (y-k)^2 = r^2$$

$h, k \rightarrow$ Centre

$x, y \rightarrow$ Point

$r \rightarrow$ Radius.

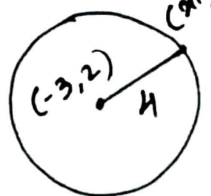
Example Find the equation of the circle with centre $(-3, 2)$ and radius (4) .

$$\text{Soln: } (x-h)^2 + (y-k)^2 = r^2$$

$$[x - (-3)]^2 + (y - 2)^2 = 4^2$$

$$(x+3)^2 + (y-2)^2 = 16$$

$$x^2 + 3^2 + 6x + y^2 + 2^2 - 4y = 16$$



$$x^2 + y^2 + 6x - 4y = 16 - 9 - 4 = x^2 + y^2 + 6x - 4y = 3$$

Example) v.v. Imp

In each of the following, find the centre and radius of the circles:

$$(x+5)^2 + (y-3)^2 = 36 \quad r=6$$

$$\Rightarrow [x - (-5)]^2 + (y-3)^2 = 6^2$$

$$(x-h)^2 + (y-k)^2 = r^2$$

$$h = -5 \mid k = 3 \mid r = 6$$

Example) In each of the following, find the centre and radius of the circles

$$x^2 + y^2 - 8x + 10y - 12 = 0$$

$$\Rightarrow x^2 - 8x + y^2 + 10y = 12$$

$$\Rightarrow x^2 - 8x + 4^2 - 4^2 + y^2 + 10y + 5^2 - 5^2 = 12$$

$$\Rightarrow x^2 - 8x + 4^2 + y^2 + 10y + 5^2 = 12 + 4^2 + 5^2$$

$$\Rightarrow (x-4)^2 + (y+5)^2 = 12 + 16 + 25$$

$$\Rightarrow (x-4)^2 + (y+5)^2 = 28 + 25$$

$$\Rightarrow (x-4)^2 + (y-(-5))^2 = 53$$

$$\Rightarrow (x-4)^2 + (y-(-5))^2 = (\sqrt{53})^2$$

$$(x-h)^2 + (y-k)^2 = r^2$$

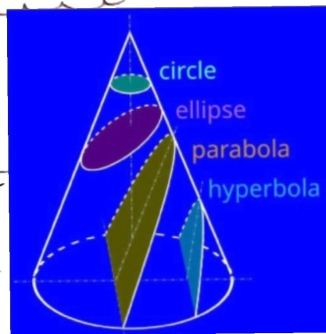
$$(h, k) = (4, -5) \quad r = \sqrt{53}$$

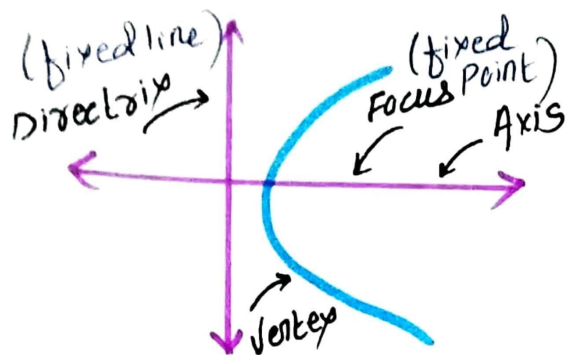
Parabola -

A Parabola is the set of all points in a plane that are equidistant from:

- a fixed point called the focus and
- a fixed line called the directrix.

It looks like the curve made when you throw a ball in the air





Important Terms:

- **Focus (F)**: The fixed Point
- **Directrix**: The fixed line
- **Axis**: A line passing through the focus and $\perp$ to the directrix
- **Vertex**: The Point where the Parabola meets the axis.

NOTE:

If the focus lies on the directrix, the curve becomes a straight line - this is called a degenerate parabola.

Degenerate case:

- Normally, the focus does not lie on the directrix.
- But if the focus lies on the directrix, then the usual parabola does not form.
- In this case, the set of equidistant points will be a straight line
- This line passes through the focus and is $\perp$ to the directrix

	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
Vertex	(0,0)	(0,0)	(0,0)	(0,0)
Focus	(a, 0)	(-a, 0)	(0, a)	(0, -a)
Directrix	$x = -a$	$x = a$	$y = -a$	$y = a$
Latus Rectum	4a	4a	4a	4a
Axis	x-axis	x-axis	y-axis	y-axis

Extra tips:

- Parabola open right/left when along x-axis (y^2 form)
- Parabolas open up/down when along y-axis (x^2 form)
- Only one squared term in the equation (either x^2 or y^2).

Latus Rectum of a Parabola:

Definition:

- The Latus Rectum is a line segment.
- Perpendicular to the axis of the parabola.
- Passes through the focus.
- Both ends lie on the parabola.

Example:

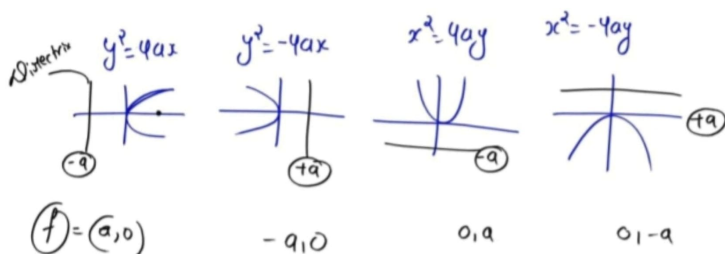
- For Parabola $y^2 = 4ax$
- Focus: (a, 0)
- Axis: x-axis
- The Latus Rectum is a vertical line through the focus.

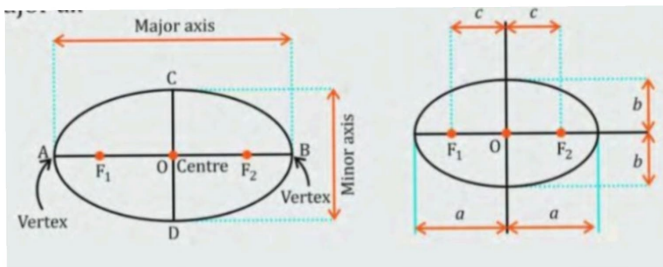
Length of Latus Rectum:

- Let's say point A & B are ends of the latus rectum.
- A and B lie above and below the focus (a, 0).
- Distance from A to F = 2a
- Distance from F to B = 2a
- So total length AB = 2a + 2a = 4a.

Final Answer -

Length of Latus Rectum = 4a





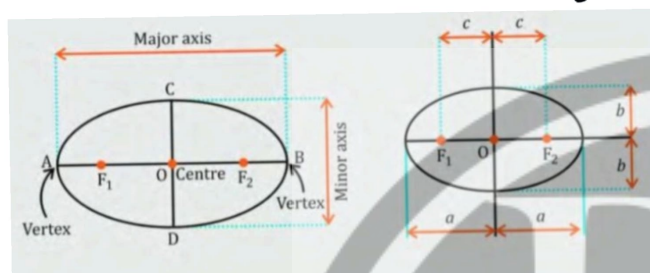
Relationship Between a , b and c :

Let -

a = semi-major axis (half of major axis)

b = semi-minor axis (half of minor axis)

c = distance from centre to focus



Eccentricity of an Ellipse:

- Eccentricity (denoted by e) is a number that tells us how "stretched" out "round" a conic section is.
- Eccentricity decides whether the conic is a circle, ellipse, parabola or hyperbola.

Eccentricity measures the shape of a conic:

- For a circle, $e = 0 \rightarrow$ Perfect Round
- For an ellipse, $0 < e < 1 \rightarrow$ oval shape
- For a parabola, $e = 1 \rightarrow$ Open curve
- For a hyperbola, $e > 1 \rightarrow$ Two separate open curves.

Eccentricity of an Ellipse -

Definition:

The eccentricity (e) of an ellipse is the ratio:

$$e = \frac{c}{a} \Rightarrow c = ae$$

Where,

c = dist from centre to focus

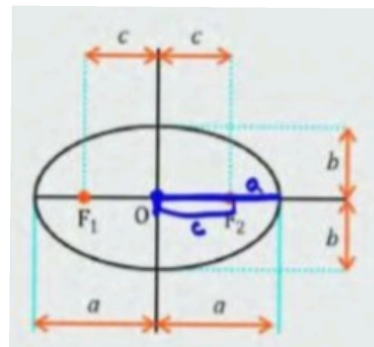
a = dist from centre to vertex (semi-major axis)

Key points:

For an ellipse

$$0 < e < 1$$

(This means the focus lies inside the ellipse, not at the end).



In terms of e :

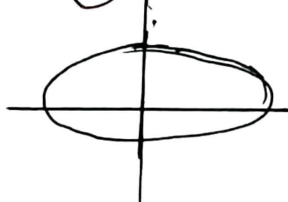
Since $c = ae$, the focus is at a distance of ae from the centre.

	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1; a > b$	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1; a > b$
Equation	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$
Centre	$(0, 0)$	$(0, 0)$
e = eccentricity	$e = \frac{\sqrt{a^2 - b^2}}{a}$	$e = \frac{\sqrt{a^2 - b^2}}{a}$
Foci	$(\pm ae, 0)$	$(0, \pm ae)$
Latus Rectum	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$
Length of major Axis	$2a$	$2a$
Length of minor Axis	$2b$	$2b$
Vertices	$(\pm a, 0)$	$(0, \pm a)$

X-axis

$$\frac{x^2}{6^2} + \frac{y^2}{4^2} = 1$$

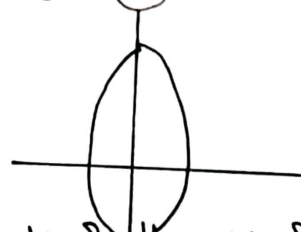
$$e = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a} = \frac{\sqrt{36 - 16}}{6} = \frac{\sqrt{20}}{6}$$



Y-axis

$$\frac{x^2}{4^2} + \frac{y^2}{6^2} = 1$$

$$e = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a} = \frac{\sqrt{36 - 16}}{6} = \frac{\sqrt{20}}{6}$$



QUEST) In each of the, find the coordinate of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

$$\frac{x^2}{4} + \frac{y^2}{25} = 1$$

$$\frac{x^2}{2^2} + \frac{y^2}{5^2} = 1$$

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

$$a=5, b=2$$

Eccentricity: Foci:

$$e = \frac{\sqrt{a^2 - b^2}}{a}$$

$$e = \frac{\sqrt{25-4}}{5}$$

$$e = \frac{\sqrt{21}}{5}$$

$(0, \pm ae)$

$$[0, \pm 5(\frac{\sqrt{21}}{5})]$$

$$\Rightarrow (0, \pm \sqrt{21})$$

Vertices:

$$(0, \pm a)$$

$$(0, \pm 5)$$

* Length $\rightarrow$ Major axis $2a$
 $2(5)$
 10

* Length $\rightarrow$ Minor axis $2b$
 $2(2)$
 4

Locus Rectum -

$$e \rightarrow \frac{2b^2}{a} = \frac{2(2)^2}{5} = \frac{8}{5}$$

$$\text{Vertical Ellipse } \frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

Question) Centre at $(0,0)$, major axis on the y-axis and passes through the point $(3,2)$ and $(1,6)$.

Pass $(1,6)$ Pass $(3,2)$

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

$$\frac{3^2}{b^2} + \frac{2^2}{a^2} = 1$$

$$e \rightarrow \frac{(1)^2}{b^2} + \frac{(6)^2}{a^2} = 1$$

$$\frac{9}{b^2} + \frac{4}{a^2} = 1 \quad \text{--- (ii)}$$

$$\frac{1}{b^2} + \frac{36}{a^2} = 1 \quad \text{--- (i)}$$

Eqn (i) & Eqn (ii):

$$\frac{1}{b^2} + \frac{36}{a^2} = 1$$

$$\frac{81}{b^2} + \frac{36}{a^2} = 1$$

$$(-) \quad (-) \quad (-)$$

$$\frac{1}{b^2} - \frac{81}{b^2} = 8$$

$$80 = 8b^2$$

$$b^2 = 10$$

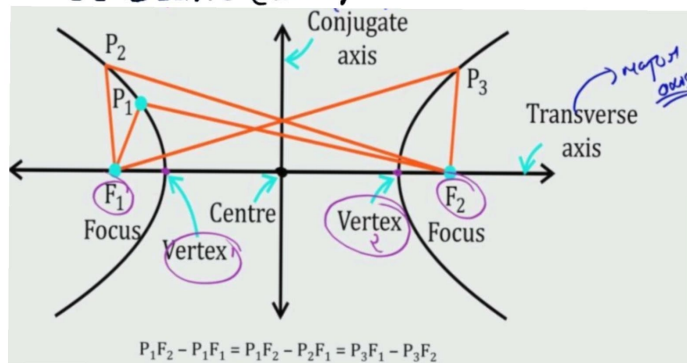
Hyperbola -

A Hyperbola is the set of all points in a plane where the difference of distance from

two fixed point (called foci) is constant.

Think:

Farther distance - closer distance = constant ($2a$)



Important Terms:

Term	Meaning
Foci	Two fixed points (F_1 and F_2)
Centre	Midpoint of F_1F_2
Transverse Axis	Line joining the foci
Conjugate Axis	Line perpendicular to the transverse axis through the centre
Vertices	Points where the hyperbola cuts the transverse axis
$2a$	Distance between the vertices
$2c$	Distance between the foci
b	Given by $b = \sqrt{c^2 - a^2}$

Eccentricity of Hyperbola -

- Formula: $e = c/a$
- Since $c > a$, eccentricity $e > 1$
- In terms of e : $c = ae$

$$\text{C/O/A V.V. Imp} \\ \text{(Relation) } c^2 = a^2 + b^2$$

Quest) In each following, find the equations of the hyperbola satisfy the given conditions vertices $(\pm 2, 0)$, foci $(\pm 3, 0)$.

$$\rightarrow (\pm a, 0) \rightarrow (\pm c, 0)$$

$$\rightarrow a = 2 \rightarrow c = 3$$

W.K.T

$$c^2 = a^2 + b^2$$

$$3^2 = 2^2 + b^2$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{2^2} - \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{4} - \frac{y^2}{5} = 1$$

final equation.

$$9 = 4 + b^2$$

$$9 - 4 = b^2$$

$$5 = b^2$$

$$b = \pm\sqrt{5}$$

KBD

(#) $(\pm a, 0)$ (#) $(0, \pm b)$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

अलग से

Transverse Axis = $2a = 2 \times 4 = 8$

Conjugate Axis = $2b = 2 \times 3 = 6$

Quest) In the quest foci $(\pm 5, 0)$, the transverse axis is of length 8. Find Eqⁿ of Hyperbola.

Foci: $(\pm c, 0)$

$\hookrightarrow c = 5$

$\therefore \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

WKT:

$$c^2 = a^2 + b^2$$

$$25 = 16 + b^2$$

$$b = 3$$

$$2a = 8$$

$$a = 4$$

$$a = 4$$

$$b = 3$$

Ans

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

Quest) In each of the, find the co-ordinates of the foci and the vertices, the eccentricity and the length of the latus rectum of the hyperbolas $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

Ellipse/Hyperbola

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{4^2} - \frac{y^2}{3^2} = 1$$

compare

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Foci: $(\pm c, 0)$ or $(\pm ae, 0)$

$$a = 4, b = 3$$

WKT: $c^2 = a^2 + b^2$

$$c^2 = 16 + 9$$

$$c^2 = 25$$

$$c = \sqrt{25}$$

$$c = 5$$

Foci = $(\pm 5, 0)$

Vertices:

$\hookrightarrow (\pm a, 0)$

$\hookrightarrow v = (\pm 4, 0)$

Latus Rectum = $\frac{2b^2}{a}$

$$l = \frac{2 \times 9}{4}$$

$$l = 9/2$$

eccentricity (e):

$$e = \frac{\sqrt{a^2 + b^2}}{a}$$

$$e = \frac{\sqrt{16 + 9}}{4}$$

$$e = \frac{5}{4}$$

Quest) Find the Eqⁿ of Hyperbola. Foci $(0, \pm\sqrt{10})$ Passing through $(2, 3)$.

$\hookrightarrow (0, \pm c)$

$$c = \sqrt{10}$$

WKT:

$$c^2 = a^2 + b^2$$

$$(\sqrt{10})^2 = a^2 + b^2$$

$$10 = a^2 + b^2$$

$$10 - a^2 = b^2 \quad \text{--- (1)}$$

NOW,

$$\Rightarrow \frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$\Rightarrow \frac{3^2}{a^2} - \frac{2^2}{b^2} = 1$$

$$\Rightarrow \frac{9}{a^2} - \frac{4}{10 - a^2} = 1$$

$$\Rightarrow \frac{9(10 - a^2) - 4a^2}{a^2(10 - a^2)} = 1$$

$$\Rightarrow \frac{90 - 9a^2 - 4a^2}{10a^2 - a^4} = 1$$

$$\Rightarrow 90 - 13a^2 = 10a^2 - a^4$$

From Eqⁿ (1)

$$10 - a^2 = b^2$$

$$10 - 18 = b^2$$

$$-8 = b^2$$

$$b = \pm\sqrt{-8}$$

Not Possible

$$10 - 5 = b^2$$

$$5 = b^2$$

$$b = \pm\sqrt{5}$$

$$\Rightarrow a^4 - 23a^2 + 90 = 0$$

$$\text{Let } a^2 = t$$

$$\Rightarrow t^2 - 23t + 90 = 0$$

$$\Rightarrow t^2 - 18t - 5t + 90 = 0$$

$$\Rightarrow t(t - 18) - 5(t - 18) = 0$$

$$\Rightarrow t(t - 18) - 5(t - 18) = 0$$

$$\Rightarrow t(t - 18) - 5(t - 18) = 0$$

$$\Rightarrow t(t - 18) - 5(t - 18) = 0$$

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